

MEASURING ANONYMITY IN COMPLEX NETWORKS

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INTRODUCTION

Researching networks can lead to a variety of interesting insights. Before sharing this sensitive research, however, entities represented should be *anonymous* to avoid a potential breach in privacy. In the field of Statistical Disclosure Control (SDC) anonymization of tabular data is researched extensively [1], but here network data is yet uncharted territory.

PROBLEM & APPROACH

When choosing an anonymization measure, one should first take into account how much information can be obtained by an attacker. Second, one should make a trade-off between *anonymity* and *data-utility*.

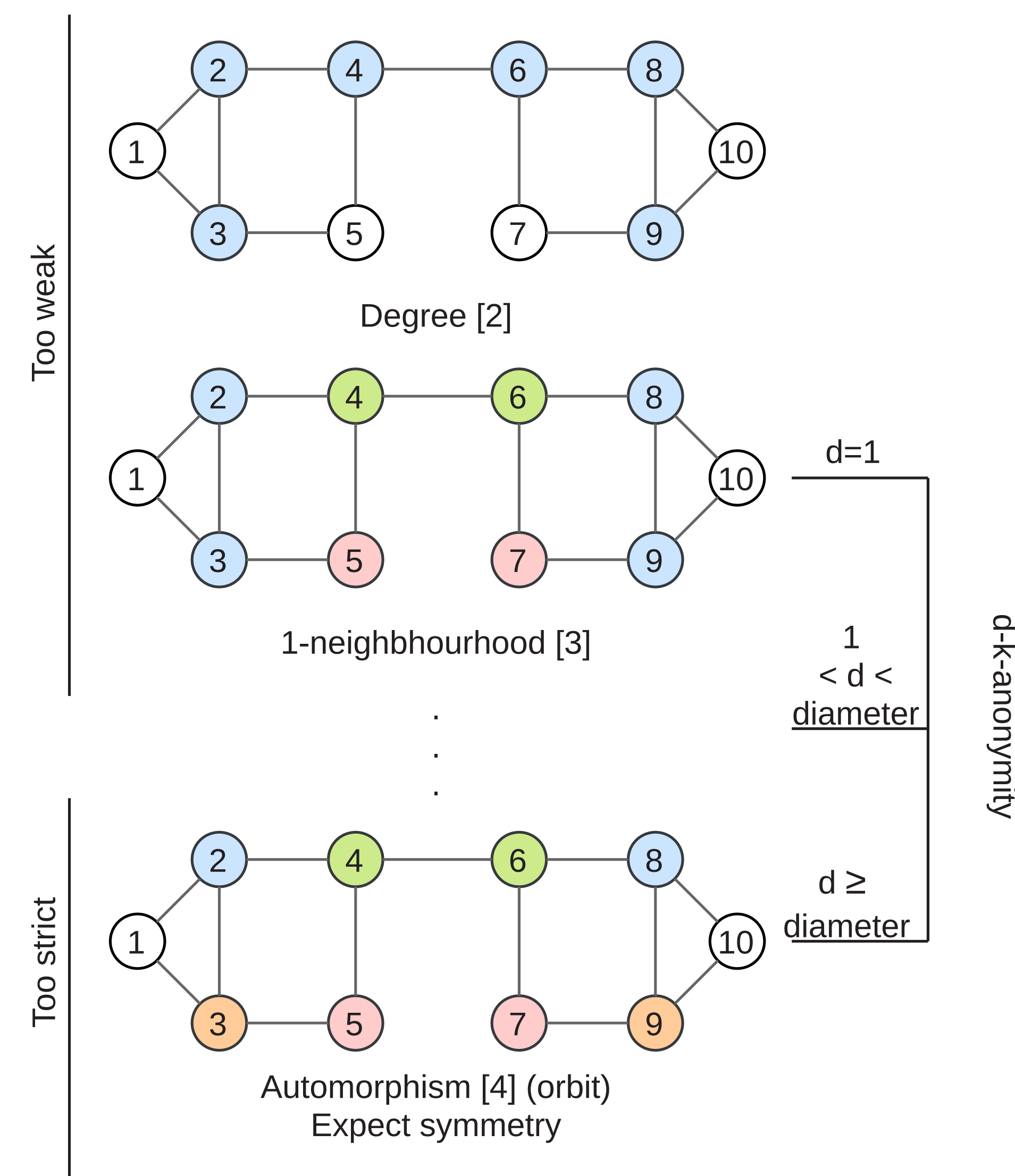
However, most existing measures are often too weak, such as degree [2] or 1-neighbourhood isomorphism [3], or too strict; for example k -automorphism [4] which expects symmetry in the graph.

Therefore we introduce the parametrized measure **d - k -anonymity**. This can behave as 1-neighbourhood isomorphism and k -anonymity, and approach the latter. Here two vertices v_1 and v_2 are equivalent if:

- Their respective d -neighbourhoods are isomorphic
- There is at least one isomorphism mapping v_1 onto v_2

MAIN CONTRIBUTIONS

- A new parametrized measure for anonymity that **balances anonymity and data-utility**: d - k -anonymity
- We present results on well-known random graph models
- We can measure the anonymity of the full 15-million node **family network of the Netherlands**



RESULTS: GRAPH MODELS

By using Nauty [5] and heuristics, runtimes were obtained of: ≈ 2 hrs. on the family network of the Netherlands and <60 seconds on ER and BA graphs with 10,000 vertices.

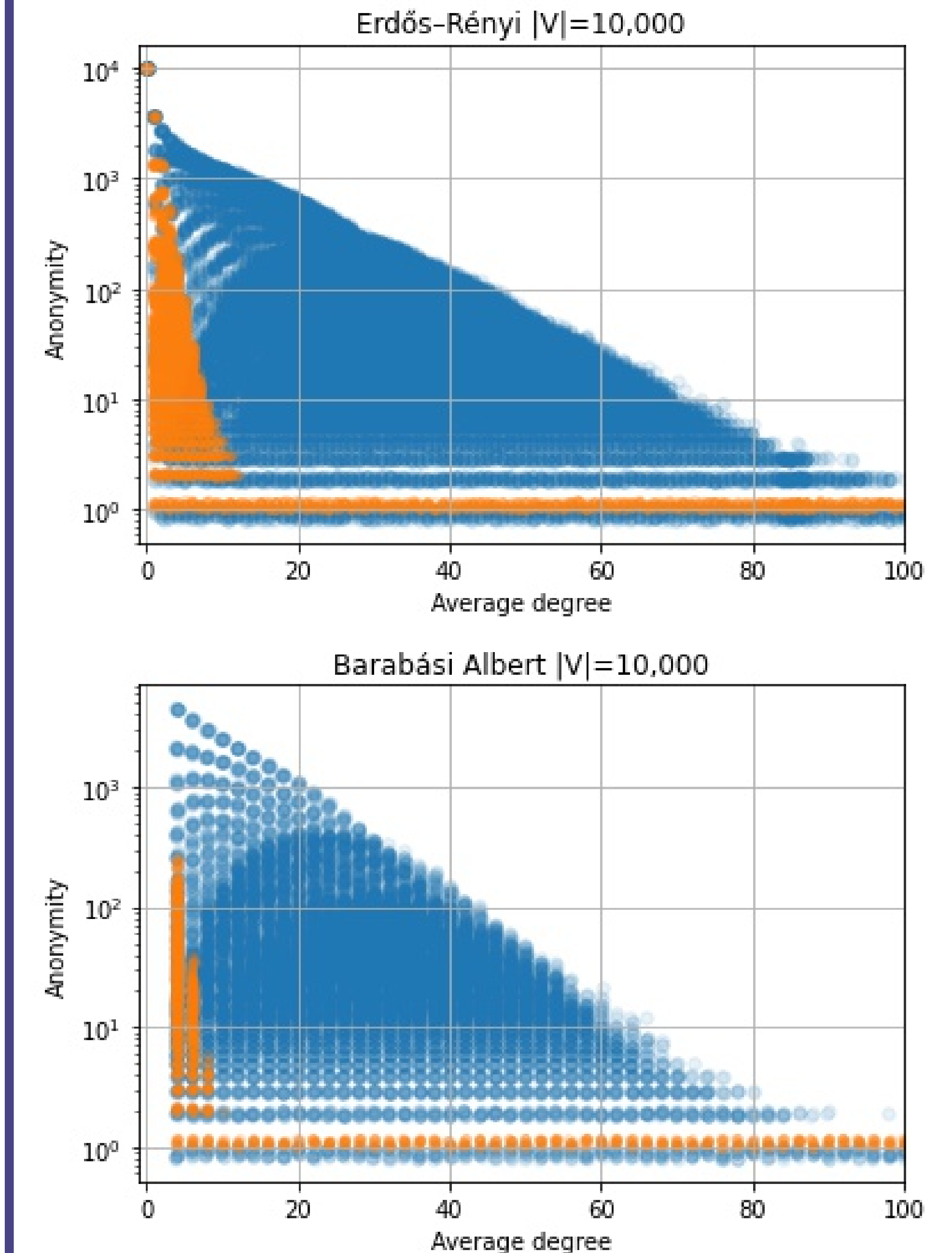


Figure 1: Anonymity distribution on ER and BA graphs w.r.t 1- k -anonymity ($d=1$) and 2- k -anonymity ($d=2$). Each dot represents a group of equivalent vertices. **Most anonymity is lost at $d=2$.**

RESULTS: FAMILY NETWORK OF THE NETHERLANDS

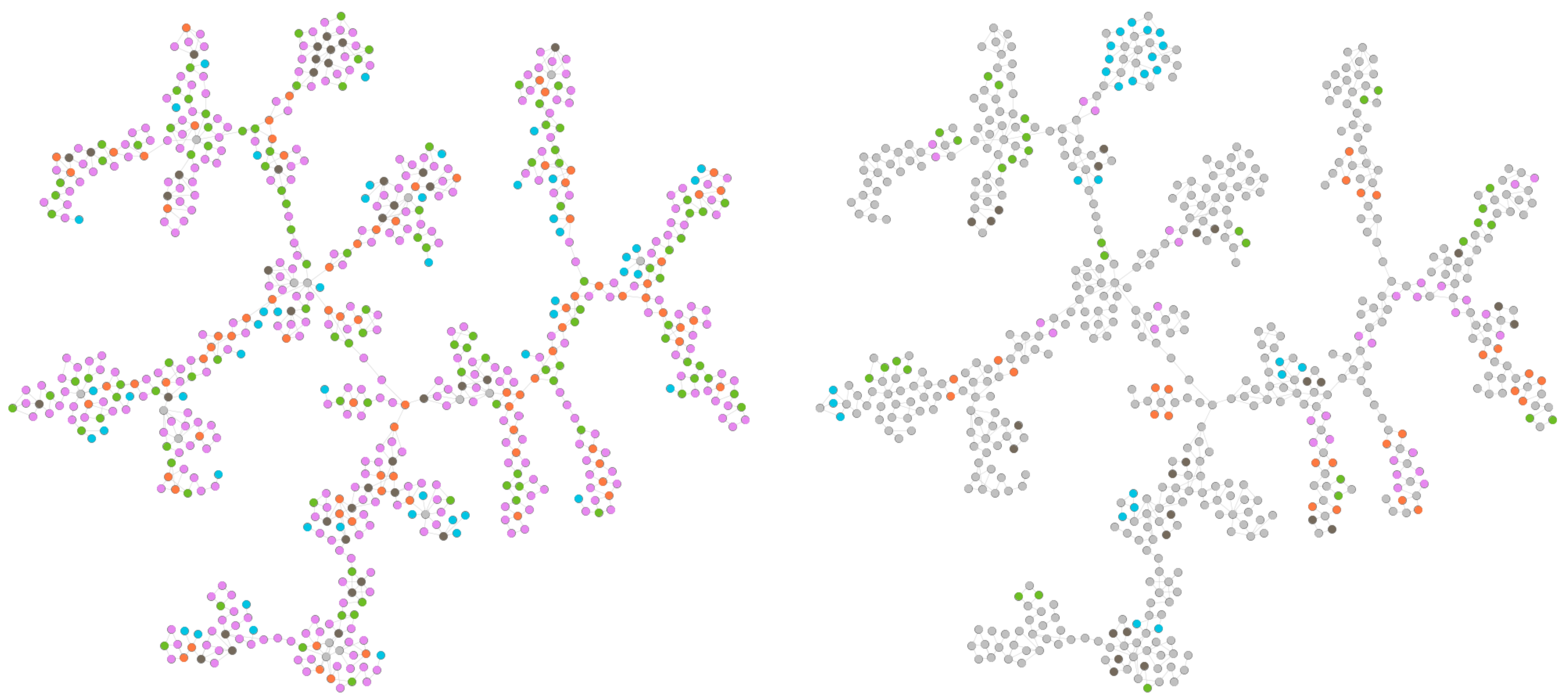


Figure 2: Part of dutch family Network, edges represent a parent-child relation. 5 largest groups w.r.t. 1- (left) and 2- k -anonymity (right) are coloured.

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